

PAPER_1490 — UQFF: $2\pi/13.8$ Hubble Oscillation per Gyr (Bucket C)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket C — derived from F:/Aetheric Propulsion source material **Date:** June 16, 2026 **Location:** 41.0997 N, 80.6495 W (Youngstown, OH, USA) **Status:** CLOSED — $\omega_{\text{Hubble}} = 2\pi/13.8 \text{ Gyr}^{-1} = 0.4553 \text{ rad/Gyr}$

Observation

Master gravity equations include a $2\pi/13.8$ oscillation factor referencing the 13.8 Gyr universe age as the structural period.

UQFF Closed Identity

$\omega_{\text{Hubble}} = 2\pi / t_{\text{universe}} = 2\pi / 13.8 \text{ Gyr} = 0.4553 \text{ rad/Gyr}.$

Physical Interpretation

Universal time-oscillation factor in every master gravity equation. The 13.8 Gyr universe age sets the structural period; the 2π factor encodes the $F_U=1$ ledger phase wrap. Appears in Magnetar, Sgr A*, AGN, and stellar master equations alike.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical UQFF integer primitives $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$. Zero free parameters.

Reference

- Source: F:/Aetheric Propulsion (Daniel's reactor + experimental docs, 2025-2026)
 - Related: PAPER_646 (Universal Inertial Operator, Caduceus, DPM mechanism), PAPER_872 (proto-element nuclear identity), PAPER_087 (BH thermodynamics), PAPER_1156 (cosmology suite), PAPER_1255 (muonic hydrogen radius).
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